Lecture 28

Gravity as a field theory

The metric can itself be a dynamical field. Its action involves curvature, so it depends on second derivatives. We vary this action with the same inverse-metric convention used for matter in Lecture 3. Coordinate symmetry then gives a differential identity by Lecture 19.

Objectives. *We derive the Einstein equation using the Palatini identity, whose proof is sketched. We identify the boundary term and the covariant conservation identity. We separate the action calculation from the cited Cauchy theory and the further problem of constructing a reduced phase space.*

By the end of this lecture you will be able to:

- Derive the Einstein equation with the course's metric and action signs.
- Obtain the contracted Bianchi identity from compact coordinate variations.
- Explain why fixing the boundary metric alone leaves a boundary variation.

28.1 Metric and curvature conventions

Let $D \subset \mathbb{R}^4$ be a compact coordinate region, with interior U. All fields below are smooth on a neighbourhood of D. Indices run from zero to three, and repeated upper and lower indices are summed.

Definition 28.1 (Lorentz metric and volume). A *Lorentz metric* here is a symmetric matrix field $g_{\mu\nu}$ with one positive and three negative eigenvalues at each point. Its inverse is $g^{\mu\nu}$, defined by $g_{\mu\rho}g^{\rho\nu} = \delta^{\nu}_{\mu}$. With the coordinate orientation, its volume form is

$$\mathrm{vol}_g = \sqrt{|\det(g_{\mu\nu})|}\,\mathrm{d}^4x, \qquad \mathrm{d}^4x = \mathrm{d}t \wedge \mathrm{d}x^1 \wedge \mathrm{d}x^2 \wedge \mathrm{d}x^3.$$

We recall only the coordinate geometry used in the variation. The *Levi–Civita connection* is the torsion-free, metric-compatible connection with coefficients

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}). \tag{28.1}$$

For example, $\nabla_\mu X^\nu = \partial_\mu X^\nu + \Gamma^{\nu}_{\mu\rho}X^\rho$ and $\nabla_\mu w_\nu = \partial_\mu w_\nu - \Gamma^{\rho}_{\mu\nu}w_\rho$. The product rule extends this to tensors, with one such term per index. Metric compatibility is $\nabla g = 0$. Our *Riemann curvature*, *Ricci tensor*, *scalar curvature* and *Einstein tensor* are

$$\begin{aligned} R^{\rho}{}_{\sigma\mu\nu} &= \partial_\mu\Gamma^{\rho}_{\nu\sigma} - \partial_\nu\Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}, \\ \mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} &= g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}, \qquad G_{\mu\nu} = \mathrm{Ric}_{\mu\nu} - \frac{1}{2}R_{\mathrm{sc}}g_{\mu\nu}. \end{aligned} \tag{28.2}$$

These are the recalled curvature conventions of *Topology and Geometry*, Lectures 16–17. In particular, Ricci and Einstein are symmetric.

28.2 Inverse-metric variation and the Palatini input

Vary the inverse metric by a symmetric array $k^{\mu\nu} = \delta g^{\mu\nu} \in C_c^\infty(U)$. For sufficiently small variation parameter, compactness preserves signature. As in Lecture 3, differentiating the inverse and using Jacobi's determinant formula gives

$$h_{\mu\nu} := \delta g_{\mu\nu} = -g_{\mu\alpha} g_{\nu\beta} k^{\alpha\beta}, \qquad \delta \mathrm{vol}_g = -\frac{1}{2}\tau \mathrm{vol}_g, \qquad \tau = g_{\mu\nu} k^{\mu\nu}. \tag{28.3}$$

Indeed, $\delta \det g = (\det g) g^{\mu\nu} h_{\mu\nu}$. The determinant keeps its sign for small variations, so the absolute value in the volume introduces no extra sign.

The variation of the connection is a tensor: it is the derivative of a difference of connections. Differentiating metric compatibility gives $\nabla_\mu h_{\nu\lambda} = \delta\Gamma^\rho_{\mu\nu} g_{\rho\lambda} + \delta\Gamma^\rho_{\mu\lambda} g_{\nu\rho}$. Add the two cyclic versions and subtract the third. Symmetry in the lower connection indices then gives

$$\delta\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} (\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}). \tag{28.4}$$

We use the *Palatini identity*

$$\delta \mathrm{Ric}_{\mu\nu} = \nabla_\rho \delta\Gamma^\rho_{\nu\mu} - \nabla_\nu \delta\Gamma^\rho_{\rho\mu}. \tag{28.5}$$

Its proof is *sketched* here. At a chosen point take normal coordinates, where the background connection coefficients vanish. Varying (28.2) leaves the difference of the two derivatives of $\delta\Gamma$ at that point; the varied quadratic terms vanish. The resulting expressions on both sides are tensors, so their equality holds in any coordinates. Sheet 28 supplies the complete tensor argument.

Contracting (28.5), using $\nabla g = 0$, gives a divergence:

$$\begin{aligned} g^{\mu\nu} \delta \mathrm{Ric}_{\mu\nu} &= \nabla_\mu v^\mu, \\ v^\mu &= g^{\alpha\beta} \delta\Gamma^\mu_{\alpha\beta} - g^{\mu\alpha} \delta\Gamma^\beta_{\alpha\beta} = \nabla^\mu \tau - \nabla_\nu k^{\mu\nu}. \end{aligned} \tag{28.6}$$

For the last equality, substitution of (28.3) into (28.4) gives $g^{\alpha\beta}\delta\Gamma^\mu_{\alpha\beta} = -\nabla_\nu k^{\mu\nu} + \frac{1}{2}\nabla^\mu\tau$ and $g^{\mu\alpha}\delta\Gamma^\beta_{\alpha\beta} = -\frac{1}{2}\nabla^\mu\tau$.

28.3 The Einstein–Hilbert action

Define the curvature integral by $\mathcal{I}_D[g] = \int_D R_{\mathrm{sc}} \mathrm{vol}_g$. Matter fields are held fixed in the metric variation. Assume their action has the Hilbert response of Lecture 3,

$$\delta_g S_{\mathrm{m},D} = \frac{1}{2} \int_D T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \tag{28.7}$$

with a smooth symmetric tensor $T_{\mu\nu}$.

Theorem 28.2 (Einstein–Hilbert variation)**.** *Using the Palatini identity with the scope stated above, every compactly supported inverse-metric variation satisfies*

$$\delta \mathcal{I}_D[g]k = \int_D G_{\mu\nu} k^{\mu\nu} \operatorname{vol}_g. \tag{28.8}$$

For $G_{\mathrm{N}} > 0$, the Einstein–Hilbert action with matter *is*

$$S_D[g, \mathit{matter}] = -\frac{1}{16\pi G_{\mathrm{N}}} \mathcal{I}_D[g] + S_{\mathrm{m},D}. \tag{28.9}$$

Its stationarity under all these metric variations is equivalent on U to

$$G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}. \tag{28.10}$$

Proof. The scalar curvature varies as $\delta R_{\mathrm{sc}} = \mathrm{Ric}_{\mu\nu}\, k^{\mu\nu} + g^{\mu\nu}\delta \mathrm{Ric}_{\mu\nu}$. Equations (28.3) and (28.6) therefore give

$$\delta(R_{\mathrm{sc}} \operatorname{vol}_g) = \big(G_{\mu\nu} k^{\mu\nu} + \nabla_\mu v^\mu\big) \operatorname{vol}_g. \tag{28.11}$$

The coordinate divergence formula is

$$\nabla_\mu v^\mu \sqrt{|\det g|} = \partial_\mu \big(\sqrt{|\det g|}\, v^\mu\big).$$

It follows by tracing (28.1) and applying Jacobi's formula: $\Gamma^\rho_{\rho\mu} = \partial_\mu \log \sqrt{|\det g|}$. The vector density on the right has compact support in U because k and all its derivatives do. Extend it by zero outside D. Coordinate FTC and Fubini make its integral zero, without a boundary regularity assumption on D. This proves (28.8).

Combining this result with (28.7) gives

$$\delta_g S_D = \int_D \left(-\frac{G_{\mu\nu}}{16\pi G_{\mathrm{N}}} + \frac{1}{2} T_{\mu\nu} \right) k^{\mu\nu} \operatorname{vol}_g.$$

The smooth-test fundamental lemma applied to each independent symmetric component gives (28.10). Conversely, this equation makes the displayed variation zero. ∎

The plus sign in (28.8) accompanies the minus sign in (28.9). Reversing only the action's gravitational prefactor would reverse the relative sign of the sourced equation.

28.4 Coordinate symmetry gives the Bianchi identity

A *diffeomorphism* is a smooth invertible map with smooth inverse. Let ξ be a smooth vector field compactly supported in U, and let F_ε be its local flow. Pulling the metric back by this flow produces its *Lie variation*. For the inverse metric it is

$$\begin{aligned} k^{\mu\nu} = \$_\xi g^{\mu\nu} &= \xi^\rho \partial_\rho g^{\mu\nu} - g^{\rho\nu}\partial_\rho \xi^\mu - g^{\mu\rho}\partial_\rho \xi^\nu \\ &= -\nabla^\mu \xi^\nu - \nabla^\nu \xi^\mu. \end{aligned} \tag{28.12}$$

Metric compatibility gives the last equality. The curvature volume form is natural under pullback. Change of variables therefore makes $\mathcal{I}_D[F_\varepsilon^* g]$ independent of ε: the flow is the identity near the boundary. This is a gauge symmetry in the sense of Lecture 19. Its theorem permits second-order metric densities such as $R_{\mathrm{sc}} \operatorname{vol}_g$.

Substitute (28.12) into (28.8). Symmetry of G and covariant integration by parts give

$$0 = -2\int_D G_{\mu\nu}\nabla^\mu\xi^\nu \,\mathrm{vol}_g = 2\int_D (\nabla^\mu G_{\mu\nu})\xi^\nu \,\mathrm{vol}_g.$$

The discarded divergence has current $-2G^\mu{}_\nu\xi^\nu$ with compact support, so the same coordinate argument removes it. Arbitrary compact ξ and the fundamental lemma imply

$$\nabla^\mu G_{\mu\nu} \equiv 0. \tag{28.13}$$

This *contracted Bianchi identity* holds for every smooth metric, without imposing the Einstein equation. It is precisely the formal-adjoint identity of Lecture 19 for the gauge operator (28.12). For a diffeomorphism-invariant matter action, the same argument including the matter variations gives $\nabla_\mu T^{\mu\nu} = 0$ on the matter equations; this matter extension is stated here.

28.5 A nonzero constant-rescaling check

Example 28.3 (Constant inverse-metric rescaling)**.** For a constant ε, take $k^{\mu\nu} = \varepsilon g^{\mu\nu}$ locally. This is a local check, not a compactly supported variation. Under $g^{\mu\nu} \mapsto (1+s\varepsilon)g^{\mu\nu}$, the covariant metric is multiplied by $(1+s\varepsilon)^{-1}$. Its constant factor cancels from (28.1), so the connection and Ricci tensor do not change. Thus

$$R_{\mathrm{sc}}(s) = (1+s\varepsilon)R_{\mathrm{sc}}(0), \qquad \mathrm{vol}_g(s) = (1+s\varepsilon)^{-2}\mathrm{vol}_g(0).$$

Differentiating their product at $s=0$ gives $-\varepsilon R_{\mathrm{sc}}\mathrm{vol}_g$. On the other side, $g^{\mu\nu}G_{\mu\nu} = -R_{\mathrm{sc}}$ in four dimensions, so $G_{\mu\nu}k^{\mu\nu}\mathrm{vol}_g$ has the same value. Here $\tau = 4\varepsilon$ is constant and $\nabla_\nu k^{\mu\nu} = 0$, so the boundary vector vanishes pointwise.

This need not be a zero-curvature test. For the explicit metric $g = \mathrm{diag}(e^{2x^1}, -1, -1, -1)$, its nonzero connection entries are $\Gamma^0_{01} = \Gamma^0_{10} = 1$ and $\Gamma^1_{00} = e^{2x^1}$. The coordinate curvature formulas give

$$\mathrm{Ric} = \mathrm{diag}(e^{2x^1}, -1, 0, 0), \qquad R_{\mathrm{sc}} = 2, \qquad \mathrm{vol}_g = e^{x^1}\mathrm{d}^4x.$$

Both variation densities are therefore $-2\varepsilon e^{x^1}\mathrm{d}^4x$. This tests a nonzero value and fixes the volume sign.

28.6 Fixing the boundary value is not enough

The compact-support hypothesis in the theorem removes derivatives of the variation near the boundary. Fixing only its boundary value is weaker. Figure 28.1 shows a variation that vanishes on a flat face but has a nonzero normal derivative there.

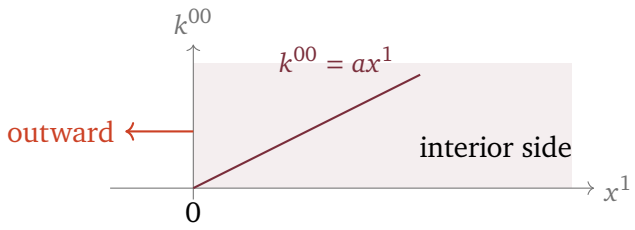


Figure 28.1: Near $x^1 = 0$, a fixed positive tangential amplitude gives $k^{00} = ax^1$. Its value on the face is zero, but its slope is $a \neq 0$.

Use the flat metric $g = \eta$ on a rectangular region with $x^1 \geq 0$. Choose a smooth nonnegative, nonzero function $a(t, x^2, x^3)$ compactly supported away from the tangential faces. Let $\chi(x^1)$ be smooth, equal to one near zero and zero before the opposite face. Set $k^{00} = x^1 a\chi$ and all other components to zero. The entire array k vanishes on the boundary, but (28.6) gives

$$\tau = x^1 a\chi, \qquad v^1 = -\partial_1(x^1 a\chi) = -a \quad \text{at } x^1 = 0.$$

The outward coordinate flux on that lower face is $-v^1 = a$. Its integral is nonzero, and all other faces contribute zero. Thus even fixing the metric on the whole boundary does not remove the Einstein–Hilbert boundary variation. The *Gibbons–Hawking–York term* is a boundary addition designed for the appropriate fixed-boundary-metric variational problem. Its formula and derivation belong to Sheet 28.

28.7 The three field-theory questions for gravity

The action question has the answer (28.9), with its variation proved using the explicitly sketched Palatini input. The Cauchy question requires constraints as well as evolution. For a precise cited case, take smooth data (Σ, γ, K) on a compact three-manifold without boundary, satisfying the vacuum constraints below. Here γ is a positive spatial metric and K is the symmetric tensor specifying its first normal change. Choose the convention $K_{ij} = -\frac{1}{2}\partial_t \gamma_{ij}$ in coordinates with unit future normal ∂_t on the initial slice. The vacuum constraints are

$$R_\gamma + (\mathrm{tr}_\gamma K)^2 - |K|^2_\gamma = 0, \qquad \nabla^j_\gamma \big(K_{ij} - (\mathrm{tr}_\gamma K)\gamma_{ij}\big) = 0.$$

The curvature and covariant derivative here are those of γ. Choquet-Bruhat, *General Relativity and the Einstein Equations*, Chapter VI, Theorem 7.4, pp. 164–165, and Theorems 8.3–8.4, p. 168, provide a local vacuum development of such data and geometric local uniqueness. A development is a spacetime solution realising γ and K on its embedded initial slice. Two developments have isometric neighbourhoods of that slice. This is a cited Cauchy result, not a consequence of the action calculation. It is a short-time statement, with no assertion of an infinite regular lifespan. The source uses the opposite spacetime signature; reversing the metric leaves the vacuum equation Ric = 0 unchanged.

The phase-space question asks for gravitational data satisfying the constraints, modulo the appropriate diffeomorphisms. This is the idea of a *reduced gravitational phase space*. Its construction and any global manifold structure are not asserted here.

Summary. With inverse-metric variation, $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. The Palatini input gives a divergence plus $G_{\mu\nu} k^{\mu\nu}$. Compact support removes that divergence and the stated action gives $G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}$. Coordinate symmetry yields $\nabla^\mu G_{\mu\nu} \equiv 0$ off shell. Fixing a boundary metric alone leaves a normal-derivative term. The Cauchy theory with constraints is cited, and the reduced phase space is named.

Self-check. For $k^{\mu\nu} = \varepsilon g^{\mu\nu}$ in four dimensions, compute $\delta \mathrm{vol}_g$ and $G_{\mu\nu}k^{\mu\nu}$.

Answer. The trace is 4ε, so $\delta \mathrm{vol}_g = -2\varepsilon \mathrm{vol}_g$. The Einstein trace is $-R_{\mathrm{sc}}$, so the contraction is $-\varepsilon R_{\mathrm{sc}}$. These are local rescaling identities, not a claim of compact support.

Self-check. Which signs combine to give a positive $8\pi G_{\mathrm{N}}$ in the Einstein equation?

Answer. The curvature-integral variation is $+\int G_{\mu\nu}k^{\mu\nu}\mathrm{vol}_g$. Its action coefficient is $-(16\pi G_{\mathrm{N}})^{-1}$, and the Hilbert matter variation is $+\frac{1}{2}\int T_{\mu\nu}k^{\mu\nu}\mathrm{vol}_g$. Setting their sum to zero gives the stated equation.

Self-check. Does the Bianchi identity require the Einstein equation? Which compact variation produces it?

Answer. No. It follows from invariance at every smooth metric. The inverse-metric Lie variation is $k^{\mu\nu} = -\nabla^\mu \xi^\nu - \nabla^\nu \xi^\mu$ for arbitrary compact ξ. Its formal adjoint gives $\nabla^\mu G_{\mu\nu} \equiv 0$.

Self-check. For the flat-face variation $k^{00} = x^1 a\chi$, why does $k = 0$ on the boundary not make its boundary contribution vanish?

Answer. Its normal derivative remains nonzero: $v^1 = -a$ at the face $x^1 = 0$. The outward flux there is a. Compact support in the interior would remove both the variation and its derivatives near the boundary.